8. Does there exist a differentiable function $f(x)$ such that $f(0)=-1, f(2)=4$ and $f'(x)\le 2$ for all x. Justify your answer.

9. Show that there lies a point on the curve $f(x)=x(x+3)e^{-\frac{\pi}{2}}, -3\le x\le 0$ where tangent drawn is parallel to the x-axis.

10. Using mean value theorem prove that for, $a>0, b>0, |e^{-a}-e^{-b}|<|a-b|$.

7.4 Series Expansions

Taylor's series and Maclaurin's series expansion of a function which are infinitely differentiable.

Theorem 7.5

(a) Taylor's Series

Let $f(x)$ be a function infinitely differentiable at $x=a$. Then $f(x)$ can be expanded as a series, in an interval $(a-\varepsilon, a+\varepsilon), \varepsilon>0$, of the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a)+\frac{f'(a)}{1!}(x-a)+\cdots+\frac{f^{(n)}(a)}{n!}(x-a)^n+\cdots.$$

(b) Maclaurin's series

If $a=0$, the expansion takes the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0)+\frac{f'(0)}{1!}x+\cdots+\frac{f^{(n)}(0)}{n!}x^n+\cdots.$$

Proof

The series expansion of $f(x)$, in powers of $(x-a)$, be given by

$$f(x) = A_0+\sum_{n=1}^{\infty} A_n(x-a)^n \qquad \ldots (7)$$

Substituting $x=a$ gives $A_0=f(a)$. Differentiation of (7) gives

$$f'(x) = 1!A_1+\sum_{n=2}^{\infty} nA_n(x-a)^{n-1} \qquad \ldots (8)$$

Substituting $x=a$ gives $A_1=f'(a)$. Differentiation of (8) gives

$$f''(x) = 2!A_2+\sum_{n=3}^{\infty} n(n-1)A_n(x-a)^{n-2} \qquad \ldots (9)$$

Substituting $x=a$ gives $A_2=\frac{f''(a)}{2!}$. Differentiation of (9) gives

$$f'''(x) = 3!A_3+\sum_{n=4}^{\infty} n(n-1)(n-2)A_n(x-a)^{n-3} \qquad \ldots (10)$$

Differentiation of (10) $(k-3)$ times gives

$$f^{(k)}(x) = k!A_k+\sum_{n=k+1}^{\infty} n(n-1)\ldots(n-k+1)A_n(x-a)^{n-k} \qquad \ldots (11)$$

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